

The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer as an anonymized observation runtime: multi-band evidence $\rightarrow$ BandPIT/WMI $\rightarrow$ a point-in-time (PIT) *world bundle*, with hard refusal when support is thin. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. $\lesssim 0.23$ under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: Compiled thin-abstain is 1.0 (100-day and full OOS $n=2000$) while Ungated soft judgment is only $\approx 0.68 / \approx 0.56$; Raw over-trades. Secondary—handoff: vintaged tilts beat momentum under a no-LLM rule ($\Delta CE=0.334$, $p=0.034$; LOBO content share 1.0). On this panel max WMI $\approx 0.093 < 0.2$, so production Compiled never opens—a valve stress test. **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Artificial intelligence**; *Machine learning*; • **Applied computing** $\rightarrow$ *Economics*.

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation** $\rightarrow$ **decision** $\rightarrow$ **execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

Failure modes. No feed $\rightarrow$ refuse (safe, useless as a data end). Thin price fragment $\rightarrow$ trade and lose (unsafe for any downstream stack). Disclosed but ungated world $\rightarrow$ read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

Table 1: End-to-end quant stack; this paper owns only the data end.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

What we build. An anonymized runtime (Section 4): multi-band evidence $\rightarrow$ BandPIT/WMI $\rightarrow$ JSON world bundle for GPT / DeepSeek / GLM / Gemini class models, served through a read API a decision layer can call. “World model” means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

How we evaluate. **RQ1 (primary—seeing):** Are ready / missing / tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (mom5), Blind. **RQ3 (secondary):** Can a transparent no-LLM rule hand off from the bundle (production refuse vs. nonempty tilts)?

Contributions. (1) Data-end semantics: epistemic observations, lag reconstruction, Π_t , and seeing vs. gating. (2) Anonymized PIT compile path: collectors $\rightarrow$ BandPIT $\rightarrow$ service surface $\rightarrow$ world bundles; frozen four-arm harness. (3) Live + handoff validation: grounding ≈ 0.97 with sufficiency dissociation; thin-abstain 1.0 at 100-day and full-OOS scale; no-LLM handoff (production refuse vs. nonempty tilts, $\Delta CE=0.334$; LOBO content share 1.0). Estimand: data-end legibility, valve reliability, and handoff usability.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream at the data end of the stack. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act. Our estimand is therefore orthogonal to memory routers and debate judges: without a typed PIT world, those stacks consume whatever fragment the tool dump provides.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality via a runtime field,

not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness. Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation Layer

Definition 3.1 (Epistemic observation). $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI); $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$. Π_t aligns clocks, gates honesty/missingness, and emits consumer JSON—the handoff to decision.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty; ACWMI adds regime weight. Compiled exposes `should_ai_abstain` and `thin_world`.

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t)$ is too low. Tilts without a gate are a dump; a gate without a readable bundle is only a switch.

Assumption 3.3 (Bounded lag and increments). Band j has max observation lag Δ_j ; latent increments satisfy $\|S_u - S_{u-1}\| \leq \tilde{\delta}$.

PROPOSITION 3.4 (LAG RECONSTRUCTION BOUND). *Under Assumption 1, $\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\tilde{\delta} + C_{\text{noi}}\epsilon_t + C_{\text{miss}}m_t$. Compilation cannot erase delay; it can refuse to pretend the bound is zero.*

Intuition. Agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.4 makes those terms explicit: a thicker raw payload can worsen reconstruction if honesty collapses. The data end therefore grades readiness and discloses gaps instead of imputing a false full world.

PROPOSITION 3.5 (COMPILATION $\neq$ DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty.*

PROPOSITION 3.6 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

PROPOSITION 3.7 (LOBO CONTENT VS. GATING). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Implication for the runtime. Props. 3.5–3.6 say the data end must export gates and abstain flags, not tilts alone: compilation without quality disclosure is still a feature dump; disclosure without a hard gate still lets thin worlds drive action. Prop. 3.7 motivates the economic check in RQ3: if deleting a durable band does not change the content rule, the bundle is empty of content.

REMARK 3.8 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface—the data end of the stack.*

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \rightarrow$ WMI; abstain flag; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / availability reads
Consumer harness	Four-arm protocol; temp. 0

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetches (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t , evidence IDs

4 Runtime: Data-End Compile Path

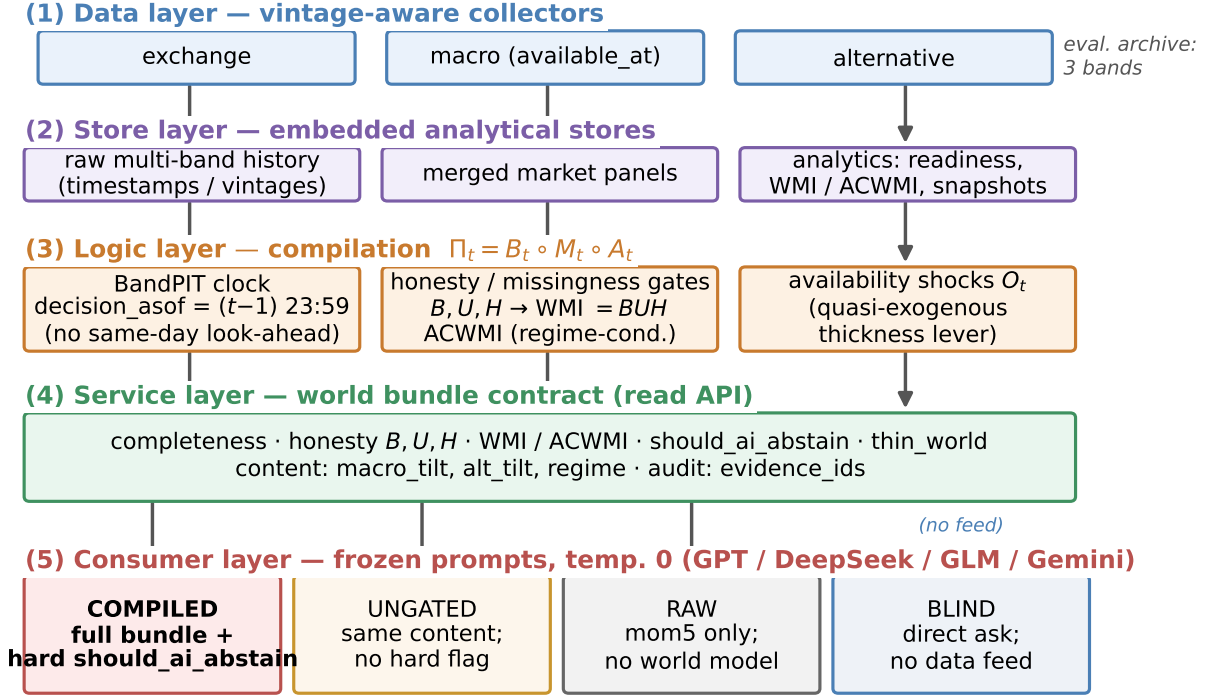
The anonymized prototype implements the data end as a layered compile path (Figure 1, Table 2). Decision and execution sit downstream; this paper does not claim them. Paths and product names are omitted for double-blind review.

4.1 Layered design

- (1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
- (2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
- (3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty/missingness gates implement Π_t ; WMI/ACWMI and availability shocks O_t (band outage events for quasi-exogenous levers) are first-class.
- (4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle and decision-bundle reads, health exposing `should_ai_abstain`, a paper world-model time slice, and availability-shock queries—the machine-readable handoff a decision layer would call.
- (5) **Consumer layer.** Frozen-prompt LLM/rule agents call an OpenAI-compatible adapter at temperature 0 under four arms.

4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of ≈ 3990 asset-days (2025-07 to 2026-08): exchange ready share ≈ 0.80 , macro 1.0, alternative 1.0. WMI is computed over the full schema, so max WMI ≈ 0.093 even on 3/3-ready days.



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

4.3 PIT clock and freshness

For calendar day t , BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the bundle records decision_asof. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.4). Breadth, stability, and honesty aggregate into $WMI_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The valve sets should_ai_abstain when $WMI < 0.2$ and exposes thin_world. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: WMI stays below the production gate throughout; ACWMI can look healthier under regime weight, so the runtime exports both and lets the configured index mode drive abstain. Table 5: production gate 0.2 never opens on this panel; $WMI \geq 0.05$ selects the band-thick subset ($\approx 61\%$ OOS).

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and evidence_ids. Listing 1 shows an exchange-gap day; Listing 2 shows a 3/3-ready day that still carries $WMI \approx 0.093$

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈ 0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

Table 5: WMI gate vs. open share (OOS). Production never opens on this panel.

Gate	Open share	Note
0.2 (production)	0.00	max WMI ≈ 0.093
0.05 (counterfactual)	0.61	$\equiv$ band-thick
0.01	1.00	near-floor open

and should_ai_abstain=true—the bundle discloses quality rather than inventing coverage.

Listing 1: Exchange-gap bundle (Compiled).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  }
}
```

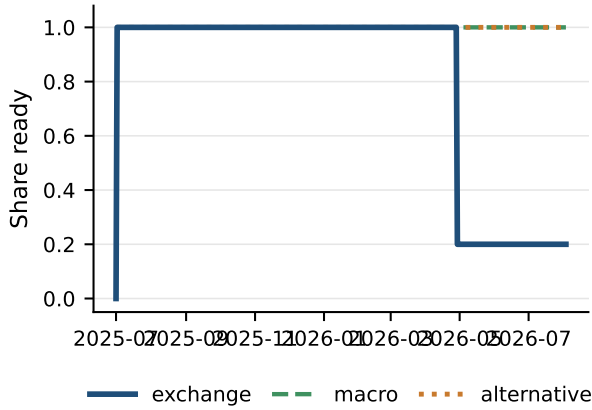


Figure 2: PIT readiness: what is visible in the archive over time.

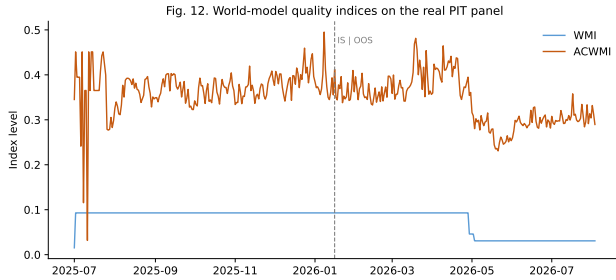


Figure 3: World-quality paths: WMI stays under the 0.2 production gate; ACWMI is regime-weighted.

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H ; main-view / stale gates
Quality	WMI, ACWMI, abstain flag, thin_world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence_ids; EAR required

```
"world_model_index":{"wmi":0.015,
"should_ai_abstain":true,"thin_world":true},
"macro_tilt":1.0,"alt_tilt":-1.0,
"audit":{"evidence_ids":[
"band:exchange:missing","band:macro:ready"]}}
```

Listing 2: 3/3-ready day still thin under schema ceiling.

```
{"date":"2026-01-16","asset":"ADA",
"completeness":{"n_ready":3,"n_missing":0},
"honesty":{"H":0.69,"U":0.38,"B_hier":0.35},
"world_model_index":{"wmi":0.093,"acwmi":0.45,
"should_ai_abstain":true,"thin_world":true},
"band_status":{"exchange":"ready","macro":"ready",
"alternative":"ready"}}
```

4.5 Four arms and frozen prompts

Actions $\in \{\text{bullish, bearish, neutral, abstain}\}$. **Compiled**: full bundle; frozen prompt *requires* abstain when should_ai_abstain is true (and when the world is judged thin). **Ungated**: same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness. **Raw**: mom5 only—thin price wiring for the ladder. **Blind**: “Today is d . How should I trade a ?” with no feed. max WMI ≈ 0.093 , so production Compiled abstains on every OOS day—a valve stress test under thin support. Compiled vs. Ungated differ only in enforcement (Listings 3–4); prompts, IS/OOS cut, and seeds are frozen. On thin Compiled days the evidence-alignment proxy (EAR) is 1.0 across vendors: rationales stay bound to disclosed evidence_ids.

Listing 3: Compiled (excerpt): hard valve.

1. If should_ai_abstain is true OR the world is thin, you MUST “abstain”.
2. Prefer band content over raw momentum (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must match evidence_ids.

Listing 4: Ungated (excerpt): soft judgment.

1. Judge whether low WMI / missingness makes action unsafe; you MAY “abstain”.
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.

Why the hard valve is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the data end’s job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.6). Ungated measures soft judgment from the same disclosed bundle; Raw measures fragment wiring; Blind measures no feed. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the valve.

4.6 Systems validation (engines)

Paper-lab smoke against the production engines returns a wired WMI/ACWMI sample with should_ai_abstain consistent with the configured index mode (WMI gate 0.2; ACWMI gate 0.35). The same indices appear in the consumer bundle; availability shocks O_t are queryable on the service surface.

5 Experiments

5.1 Setup

PIT panel ~ 399 days $\times 10$ liquid names (≈ 3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses $n=2000$; the no-LLM probe uses 199 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain rate (CE as avoided loss, not alpha); RQ3 Δ CE of the content rule vs. momentum.

Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility ≠ sufficiency chat. Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers “enough?” correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of “is this enough to trade?” is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) shows the valve contrast: Compiled abstains on should_ai_abstain; Raw goes bullish from mom5 = +1 alone.

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

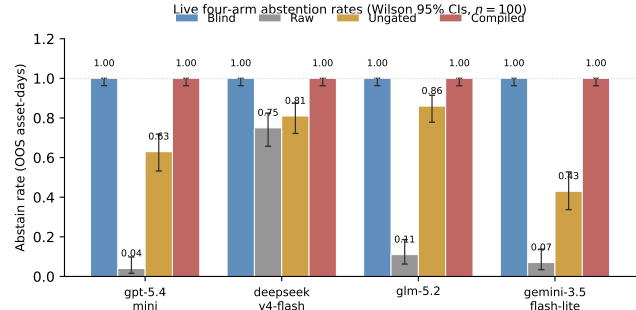
```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

Table 9: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.****Table 10: Raw action mix (100 days): half-baked context drives directional mass.**

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 9 and Figure 4: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean ΔCE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim.

Full-OOS replication. Table 11: on all 2000 OOS asset-days (every day thin under the 0.2 gate), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈0.56; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor): the failure mode that needs a data end is not “models never abstain,” but “thin fragments make them act.”

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three

Table 11: Full-OOS thin abstention ($n=2000$ asset-days; all thin under gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈ 0.56	—

*GLM coverage incomplete (Ungated $n=1402$; Raw $n=675$).**Table 12: Open slice (3/3 ready $\equiv$ WMI ≥ 0.05 ; $n \approx 1224$). Field lower bound under soft judgment.**

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	-0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	-0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	-0.25

*Raw GLM $n=398$; Ungated GLM $n=932$; others $n=1224$. Bootstrap mean CE $\approx +0.31$ ($B=5,999$ reps; one-sided $p \approx 0.21$).

of four vendors (DeepSeek -3.12, GLM -4.28, Gemini -1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

Open slice. Table 12: on WMI ≥ 0.05 ($\approx 61\%$ OOS; $n \approx 1224$) Ungated keeps compiled content without the hard boolean—mean abs. ≈ 0.48 , CE positive for every vendor (mean $\approx +0.29$; bootstrap $\approx +0.31$, CIs include 0). Production Compiled still abstains at 1.0; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). Point-positive open-slice CE and negative mixed-sample act-conditional CE can coexist: regime and act selection matter.

5.4 RQ3 (secondary): Handoff backtest—nonempty fields

The data end must be usable by a downstream decision rule without an LLM in the loop. Table 13 reads the same compiled panel under four handoff policies: production WMI gate 0.2, momentum on mom5, buy-and-hold, and a thick-ungated content rule on vintaged macro_tilt / alt_tilt. Production refuse is complete (Sharpe/CE 0, abstain 1.0); the content handoff is nonempty vs. momentum (Table 14, Figure 5): Sharpe 0.764 / CE +0.130 vs. 0.096 / -0.206; block-bootstrap $\Delta CE=0.334$ ($p=0.034$). A 2017–2026 non-vintaged audit shows no edge; a date-scramble placebo flips CE from +0.132 to -0.254, supporting timed content.

Density split and LOBO channel. Table 15: open/thick $\Delta CE=0.315$; exchange-gap days abstain more ($\Delta CE=0.129$). Table 16 and Figure 6 (Prop. 3.7): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).

5.5 Discussion

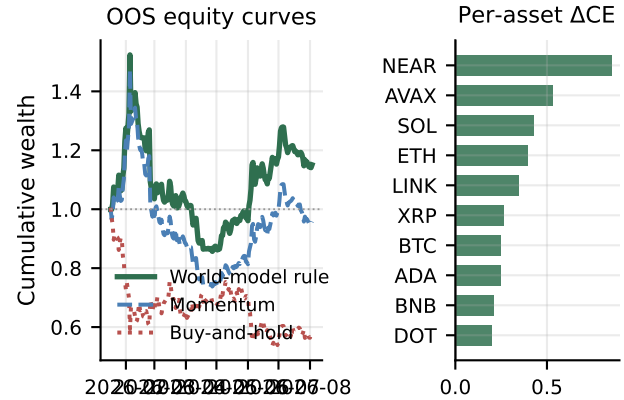
What carries the paper. (i) The data end is legible: grounding ≈ 0.97 , with sufficiency chat a poor substitute (RQ1). (ii) Thin views

Table 13: Handoff backtest on the compiled OOS panel (no LLM). Production gate vs. content rule.

Policy (reads bundle fields)	Sharpe	CE	Abstain
WMI gate 0.2 (production)	0.000	0.000	1.00
Momentum (mom5)	0.101	-0.202	0.00
Buy-and-hold	-1.399	-1.157	0.00
Thick-ungated tilts	0.767	+0.132	0.08

Table 14: No-LLM content contrast. Bootstrap ΔCE 0.334; raw gap 0.336.

Policy	Sharpe	CE
Buy-and-hold	-1.404	-1.163
Momentum	0.096	-0.206
World-bundle tilts	0.764	+0.130
Tilts – Momentum	—	0.334 ($p=0.034$)

**Figure 5: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.****Table 15: Content probe by archive density (OOS).**

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

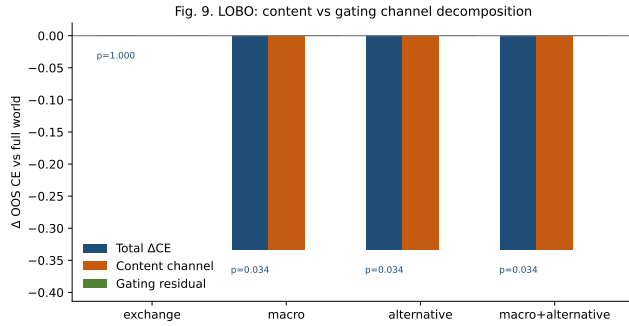
*Block-bootstrap ΔCE ; raw mean gap 0.336.

must not drive action: soft text fails at 100-day and full-OOS scale; Blind refuses with no feed; Raw acts on fragments and loses; the hard valve works (RQ2). (iii) A no-LLM handoff shows production refuse vs. nonempty content on the same bundle (RQ3); LOBO content share 1.0. Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

Why grounding first. Trading PnL confounds observation quality with policy skill. Grounding asks whether the consumer can audit

Table 16: LOBO decomposition (ungated content probe). Content share 1.0.

Drop	Δ CE content	p	Content share
macro	-0.334	0.034	1.0
alternative	-0.334	0.034	1.0
exchange	0.000	1.000	—

**Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.**

ready/missing/signed state; Raw cannot, Compiled can. The valve then asks whether soft judgment stops action on a thin disclosed world; empirically, no—at either sample size. That ordering matches the stack: a decision engine that cannot see, or that acts on thin support, is not rescued by better execution.

Design. World bundle not feature dump; previous-close PIT with explicit freshness; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. The evaluation object is the three-band bundle in Table 3; WMI is computed over the full schema so missingness stays visible. Service-layer health and bundle reads make the valve and the view callable by any downstream policy without re-ingesting collectors.

Reproducibility. Paper-lab smoke checks BandPIT/WMI/ACWMI wiring; PIT rebuild and the frozen four-arm harness regenerate the panel and live transcripts used here. Prompts, IS/OOS cut, seeds, and vendor IDs are fixed; credentials stay in environment variables. An anonymized harness ships on acceptance.

Scope. We evaluate the data end on the PIT archive above; decision and execution sit downstream.

6 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation layer: compile asynchronous evidence into a consumable PIT world bundle, disclose structural missingness, and hard-refuse when the view is too thin. Grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. Raw collapse; sufficiency chat alone fails; the hard valve abstains 1.0 on thin days (including full OOS) where soft judgment does not; vintaged tilts are nonempty (Δ CE=0.334;

LOBO content share 1.0). Build the quant data end first: compile so models can see; refuse when they must not act.

References

- [1] Chanyeol Choi, Jihoon Kwon, Alejandro Lopez-Lira, Chaewoon Kim, Minjae Kim, Juneha Hwang, Jaeseon Ha, Hojun Choi, Suyleol Yun, Yongjin Kim, and Yongjae Lee. 2025. FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 632–637. doi:10.1145/3768292.3770362
- [2] Dean Croushore and Tom Stark. 2001. A Real-Time Data Set for Macroeconomists. *Journal of Econometrics* 105, 1 (2001), 111–130.
- [3] Yitong Duan, Chuheng Zhang, and Jian Li. 2025. FactorMAD: A Multi-Agent Debate Framework Based on Large Language Models for Interpretable Stock Alpha Factor Mining. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 605–613.
- [4] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010), 1605–1641.
- [5] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates.
- [6] David Ha and Jürgen Schmidhuber. 2018. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates.
- [7] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*. OpenReview.net.
- [8] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29, 1 (2016), 5–68.
- [9] Yann LeCun. 2022. A Path Towards Autonomous Machine Intelligence. *OpenReview preprint* (2022).
- [10] Yang Li, Yangyang Yu, Haohang Li, Zhiyang Chen, and Khaldoun Khashanah. 2023. TradingGPT: Multi-Agent System with Layered Memory and Distinct Characters for Enhanced Financial Trading Performance. *arXiv preprint arXiv:2309.03736* (2023).
- [11] Yukun Liu, Aleh Tsyvinski, and Xi Wu. 2022. Common Risk Factors in Cryptocurrency. *Journal of Finance* 77, 2 (2022), 1133–1177.
- [12] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *arXiv preprint arXiv:2304.07619* (2023).
- [13] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.
- [14] Dimitris N. Politis and Joseph P. Romano. 1994. The Stationary Bootstrap. *J. Amer. Statist. Assoc.* 89, 428 (1994), 1303–1313.
- [15] Yiqing Shen, Jingshu Zhang, Feng Chen, Kaiyuan Yan, and Hongguang Li. 2025. FinSearch: A Temporal-Aware Search Agent Framework for Real-Time Financial Information Retrieval with Large Language Models. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM. doi:10.1145/3768292.3770382
- [16] Rui Sun, Zuo Bai, Wentao Zhang, Yuxiang Zhang, Li Zhao, Shan Sun, and Zhengwen Qiu. 2025. FinResearchBench: A Logic Tree based Agent-as-a-Judge Evaluation Framework for Financial Research Agents. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 656–664. doi:10.1145/3768292.3770364
- [17] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [18] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. In *FinLLM Symposium at IJCAI*.
- [19] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*. OpenReview.net.
- [20] Yangyang Yu, Haohang Li, Zhiyang Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *Proceedings of the AAAI Symposium Series* 3, 1 (2024), 595–597.
- [21] Wentao Zhang, Lingxuan Zhao, Hao Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, et al. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM.